Grade 7 Accelerated — Name___Answer Keys_______
Unit 1 — Date___________________
Lesson 3 Practice — Period__________________

Match the algebraic statement to its numeric example.

1.

Property
Division Property of Equality
Subtraction Property of Equality
Multiplication Property of Equality
Symmetric Property of Equality
Addition Property of Equality

Numeric Example
$4+5+7=9+7$
$(4+5)\times 7=9\times 7$
$(4+5)\div 7=9\div 7$
$4+5=9$
$4+5-7=9-7$

Consider the equation $33t-99=0$ to answer questions 2-5.

2. Complete the statement to describe the equation. The variable, t, is first being __multiplied___ by 33, and then 99 is being __subtracted_____ from this product. The value of the expression, $33t-99$ is ___equal__ to 0.

3. Describe the operation that could be performed to isolate $33t$ on the left side of the equation.

Add 99 to both sides. (Addition Property of equality)

4. Describe the operation that could be performed to isolate the t on the left side of the equation.

Divide both sides by 33 (Division Property of equality)

5. Solve $-4+\frac{x}{2}=16$ by applying properties of equality. Explain which properties were used to solve the equation.

$-4+\frac{x}{2}=16$ — Addition Property of Equality

$\frac{x}{2}=20$ — Multiplication Property of Equality

$x=40$

6. Solve $12n + 9 = 93$ by applying properties of equality. Explain which properties were used to solve the equation.

$$\begin{array}{rcl} 12n + 9 &=& 93 \\ \underline{-9} && \underline{-9} \\ 12n &=& 84 \\ \overline{12} && \overline{12} \end{array} \qquad n = 7$$

Use the equation $6x - 12 = 30$ to complete the following.

7. Apply properties of equality to solve the equation. Use substitution to verify the solution.

$$\begin{array}{rcl} 6x - 12 &=& 30 \\ +12 && +12 \\ 6x &=& 42 \\ \overline{6} && \overline{6} \end{array} \qquad x = 7$$

Check:

$6(7) - 12$

$42 - 12 = 30$

Use the equation $5y + 7 = 42$ to complete the following.

8. Apply properties of equality to solve the equation. Use substitution to verify the solution.

$$\begin{array}{rcl} 5y + 7 &=& 42 \\ \underline{-7} && \underline{-7} \\ 5y &=& 35 \\ \overline{5} && \overline{5} \end{array} \qquad y = 7$$

Check:

$5(7) + 7$

$35 + 7 = 42$

$42 = 42$

Use the equation $\frac{x}{5} - 8 = 17$ to complete the following.

9. Apply properties of equality to solve the equation. Use substitution to verify the solution.

$\frac{x}{5} - 8 = 17$

$\underline{+8 \quad +8}$

$5 \cdot \frac{x}{5} = 25 \cdot 5$

$x = 125$

Check:

$\frac{125}{5} - 8$

$25 - 8 = 17$

$17 = 17$

Use the equation $\frac{y}{5.2} + 3.4 = 4.3$ to complete the following.

10. Apply properties of equality to solve the equation. Use substitution to verify the solution.

$\frac{y}{5.2} + 3.4 = 4.3$

$\underline{-3.4 - 3.4}$

$5.2 \cdot \frac{y}{5.2} = 0.9 \cdot 5.2$

$y = 4.68$

Check:

$\frac{4.68}{5.2} + 3.4$

$0.9 + 3.4 = 4.3$

$4.3 = 4.3$